Assessing Learned Models for Phase-only Hologram Compression

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Abstract

Based on what I have learnt so far, I would like to systematically organize the papers, researches, classes, books that I found and explored. This paper will try to collect the recent Models, perspectives and theories in good conference or journals. In addition, it is the note to record my idea and question to improve my understanding and probably prove my directions

CCS Concepts

• **Computing methodologies → Image compression**; • **Theory of computation → Data compression**; • **Hardware →** *Displays and imagers*; *Emerging optical and photonic technologies.*

Keywords

eyes tracking, deep learning, computer graphics

ACM Reference Format:
Zicong Peng and Kaan Akşit. 2025. Assessing Learned Models for Phase-only Hologram Compression. In *Proceedings of SIGGRAPH*. ACM, New York, NY, USA, **??** pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

2 Related works

In the domains of augmented reality (AR) and physiological eye tracking, the extraction of deep physiological features via high-fidelity physical simulations and implicit neural representations has emerged as a critical research frontier. Recent studies have demonstrated the efficacy of computer graphics-based synthetic data in bridging the reality gap. This is achieved through the development of near-eye physical ray-tracing rendering pipelines [**?**] and 3D realistic reconstruction models equipped with relighting capabilities [**?**].

Building upon these advancements, deep optical features—particularly Purkinje images, which encode critical information regarding crystalline lens deformation—have been leveraged alongside machine learning and miniature multi-camera arrays to achieve highly precise tracking of dynamic accommodation and vergence [**??**]. However, capturing these subtle optical signals in real-world systems is frequently hindered by severe sampling noise [**?**]. Furthermore, the downstream requirement to drive complex hardware execution mechanisms, such as variable-focus lenses [**?**], necessitates a highly robust and mathematically pure underlying physical simulation framework.

To address these challenges, this paper proposes a novel framework utilizing Blender to construct a 3D anatomical eye model that supports complex refractive optical paths. By harnessing a rigorous optical rendering engine, this approach enables direct noise isolation and the precise extraction of high-dimensional deep features, including higher-order Purkinje reflections. Ultimately, this framework provides essential physical data priors for the design of next-generation, noise-robust perception networks and AR hardware systems.

3 Results and Discussions

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SIGGRAPH, 979-8-4007-1549-5/2025/08
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ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

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